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1. (Based on U&F, p 2.9) A 100-mm by 150-mm rectangular plate QABC is deformed into the shape shown by the dashed lines in figure 2.9. All dimensions shown in the figure are in mm. Determine at point Q:

(a) the strain components ε_x, ε_y, and γ_{xy}. Also the value for ε_{xy}; (b) the principal strains and directions of principal strains; (c) the net rotation of the deformed plate. (d) Check that the small strain assumption is valid.

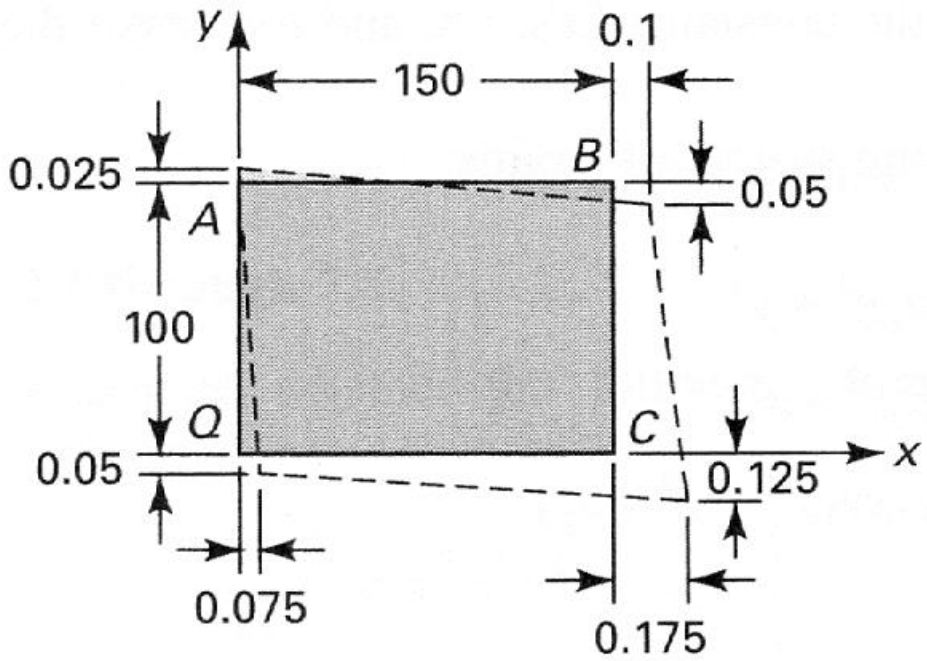


FIGURE P2.9.

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Approach to part (a)

- Define the strains at point Q using the following equations where u is a displacement along x and v is a displacement along y:

$$\varepsilon_x = \frac{\partial u}{\partial x}, \qquad \varepsilon_y = \frac{\partial v}{\partial y}, \qquad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

- The shearing strain and angular change is related by

$$\varepsilon_{xy} = \frac{1}{2}\gamma_{xy}$$

- Assuming that the strain is small, we can approximate the differentials as displacements relative to Q

$$\frac{\partial u}{\partial x} = \frac{\Delta u}{\Delta x} = \frac{u_Q - u_C}{x_Q - x_C}, \qquad \frac{\partial v}{\partial x} = \frac{\Delta v}{\Delta x} = \frac{v_Q - v_C}{x_Q - x_C}$$
$$\frac{\partial u}{\partial y} = \frac{\Delta u}{\Delta y} = \frac{u_Q - u_A}{y_Q - y_A}, \qquad \frac{\partial v}{\partial y} = \frac{\Delta v}{\Delta y} = \frac{v_Q - v_A}{y_Q - y_A}$$

Given:

- $u_Q = 0.075\ mm$
- $u_C = 0.175\ mm$
- $u_A = 0.0\ mm$
- $v_Q = -0.05\ mm$
- $v_A = 0.025\ mm$
- $v_C = -0.125\ mm$
- $x_Q = 0\ mm$
- $x_C = 150\ mm$
- $y_A = 100\ mm$
- $y_Q = 0\ mm$

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Approach to part (b)

- Using the state of strain, find the principal axis directions using

$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y}$$

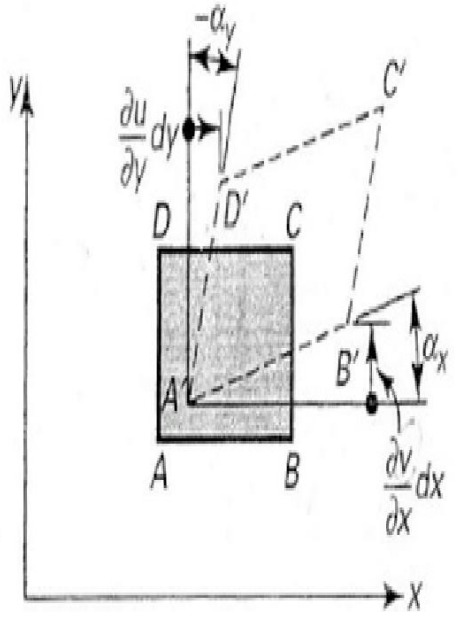
Note that there are two principal directions, and they are orthogonal to each other (separated by 180 deg on the Mohr's circle).

$$\theta_{p1} + 90^\circ = \theta_{p2}$$

- Principal strains are given by

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

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Approach to part (c)

- The textbook in section 2.3.1 defines the angle between an originally x-directed face and its deformed position as

$$\alpha_x \approx \frac{\partial v}{\partial x}$$

And for an originally y-directed face

$$\alpha_y \approx -\frac{\partial u}{\partial y}$$

The net angle of rotation is thus the average of the x and y rotations:

$$\omega = \frac{1}{2}(\alpha_x + \alpha_y)$$

Approach to part (d)

- The small strain assumption is valid if the strains are much much less than 1

$$|\varepsilon_x|, |\varepsilon_y|, |\gamma_{xy}| \ll 1$$

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Part A:

$$\frac{\partial u}{\partial x} = \frac{\Delta u}{\Delta x} = \frac{u_Q - u_C}{x_Q - x_C} = \frac{[0.075 - 0.175\ mm]}{[0 - 150\ mm]} = 0.000666$$

$$\frac{\partial v}{\partial x} = \frac{\Delta v}{\Delta x} = \frac{v_Q - v_C}{x_Q - x_C} = \frac{[-0.05\ - -0.125\ mm]}{[0 - 150\ mm]} = -0.0005$$

$$\frac{\partial u}{\partial y} = \frac{\Delta u}{\Delta y} = \frac{u_Q - u_A}{y_Q - y_A} = \frac{[0.075 - 0\ mm]}{[0 - 100\ mm]} = -0.00075$$

$$\frac{\partial v}{\partial y} = \frac{\Delta v}{\Delta y} = \frac{v_Q - v_A}{y_Q - y_A} = \frac{[-0.05\ - 0.025\ mm]}{[0 - 100\ mm]} = 0.00075$$

$$\boxed{\varepsilon_x = \frac{\partial u}{\partial x} = 0.000666}$$

$$\boxed{\varepsilon_y = \frac{\partial v}{\partial y} = 0.00075}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = [-0.00075 + -0.0005] = -0.00125$$

$$\boxed{\varepsilon_{xy} = \frac{1}{2}\gamma_{xy} = -0.000625}$$

Part B:

$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} \longrightarrow 2\theta_p = \tan^{-1}\left(\frac{-0.00125}{0.000666 - 0.00075}\right) = 86.15°$$

Since the numerator and denominator are both negative, this corresponds to a point in the 3rd quadrant, and thus

$$2\theta_p = 86.15° + 180° = 266.15° \longrightarrow \boxed{\theta_{p1} = 133.07°}$$

$$\boxed{\theta_{p2} = 133.07° + 90° = 223.07°}$$

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$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} = \frac{[0.000666 + 0.00075]}{2} \pm \sqrt{\left(\frac{[0.000666 - 0.00075]}{2}\right)^2 + ([-0.000625])^2}$$

$$\varepsilon_{1,2} = 0.000708 \pm 0.000626$$

$$\boxed{\varepsilon_1 = 0.001334} \qquad \boxed{\varepsilon_2 = 0.0000819}$$

Part C:

$$\alpha_x \approx \frac{\partial v}{\partial x} = -0.0005\ rad$$

$$\alpha_y \approx -\frac{\partial u}{\partial y} = -0.00075\ rad$$

$$\omega = \frac{1}{2}(\alpha_x + \alpha_y) = \frac{1}{2}[-0.0005 + -0.00075\ rad] = -0.000625\ rad$$

$$\boxed{\omega = -0.0358°}$$

Part D:

$$|\varepsilon_x| = 0.000666 \ll 1$$

$$|\varepsilon_y| = 0.00075 \ll 1$$

$$|\gamma_{xy}| = 0.00125 \ll 1$$

Since most of the deformation was in the +x and –y direction, it makes sense that the net rotation of the deformed plate is negative (CW). Since the normal and shear strains are all much smaller than 1 (all having magnitude around 10^-3), the small strain assumption is valid. [Time spent: 30 mins]

2. (U&F, p 2.16) A 3-m by 2-m rectangular thin plate is deformed by the movement of point B to B' as shown by the dashed lines in Fig. P2.16. Assuming a displacement field of the form $u = c_1xy$ and $v = c_2xy$, where c_1 and c_2 are dimensional constants, determine (a) expressions for the displacements based on the figure; (b) strain components ε_x, ε_y, and γ_{xy}, at point B; and (c) the normal strain $\varepsilon_{x'}$ for x' in the direction of diagonal QB. (d) Verify mathematically that the proposed strain field is physically possible. (Discuss briefly what this means.)

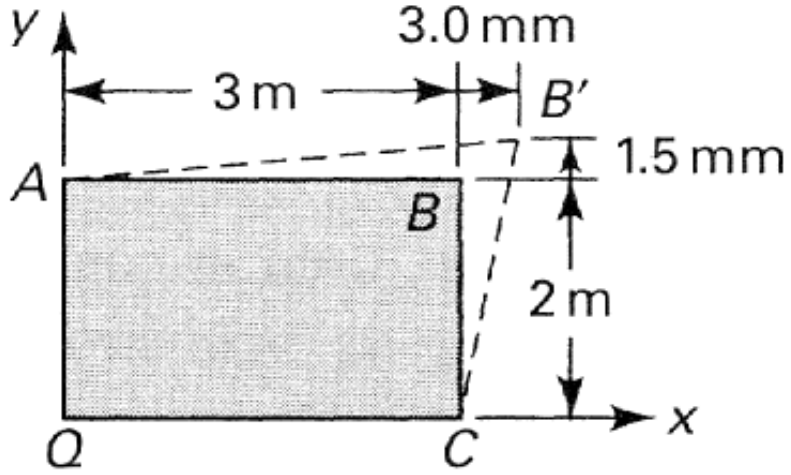


FIGURE P2.16.

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Approach to part (a)

- Solve for the constants of the displacement field using the given coordinates and position of point B

$$u(x,y) = c_1 xy$$
$$v(x,y) = c_2 xy$$

Given:

- $B = (3,2)$
- $u(3,2) = 0.003\ m$
- $\mathrm{v}(3,2) = 0.0015\ m$

Approach to part (b)

- Differentiate and evaluate at point B to solve for state of strain at B

$$\varepsilon_x = \frac{\partial u}{\partial x}, \qquad \varepsilon_y = \frac{\partial v}{\partial y}, \qquad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

Approach to part (c)

- Use 2d strain transformation to find stain along the QB direction. Start by finding the angle that the x-axis is rotated by:

$$\tan\theta = \frac{\left|\overrightarrow{BC}\right|}{\left|\overrightarrow{QC}\right|}$$

- Strain transformation

$$\varepsilon_{x'} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

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Approach to part (d)

- The derived strain field could only be physically possible if it is consistent kinematically. This is checked with the compatibility equation:

$$\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \, \partial y}$$

Part A:

$u(3,2) = c_1[3\ m][2\ m] = 0.003\ m \longrightarrow c_1 = 0.0005\ m^{-1}$

$\mathrm{v}(3,2) = c_2[3\ m][2\ m] = 0.0015\ m \longrightarrow c_2 = 0.00025\ m^{-1}$

$$\boxed{\begin{matrix} u(x,y) = 0.0005xy \\ v(x,y) = 0.00025xy \end{matrix}}$$

Part B:

$$\varepsilon_x = \frac{\partial u}{\partial x} = 0.0005y \qquad \varepsilon_x(3,2) = 0.0005[2\ m] \qquad \boxed{\varepsilon_x(3,2) = 0.001}$$

$$\varepsilon_y = \frac{\partial v}{\partial y} = 0.00025x \qquad \varepsilon_y(3,2) = 0.00025[3\ m] \qquad \boxed{\varepsilon_y(3,2) = 0.00075}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = 0.0005x + 0.00025y \qquad \gamma_{xy}(3,2) = 0.0005[3\ m] + 0.00025[2\ \mathrm{m}] \qquad \boxed{\gamma_{xy}(3,2) = 0.002}$$

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Part C:

$$\tan\theta = \frac{\left|\overrightarrow{BC}\right|}{\left|\overrightarrow{QC}\right|} = \frac{[2\ m]}{[3\ m]} \qquad \theta = 33.69^\circ$$

$$\varepsilon_{x'} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\varepsilon_{x'} = \frac{[0.001 + 0.00075]}{2} + \frac{[0.001 - 0.00075]}{2}\cos(2 \cdot 33.69^\circ) + \frac{[0.002]}{2}\sin(2 \cdot 33.69^\circ)$$

$$\boxed{\varepsilon_{x'} = 0.00447}$$

Part D:

$$\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x\, \partial y} \longrightarrow 0 + 0 = 0$$

Compatibility is satisfied so the strain field is physically possible. This means that the deformed plate would remain a continuous material without any gaps. It also makes sense that the strain along the x' direction would be the greatest since point B is displaced upward and to the right, causing the diagonal to elongate. [Time spent: 30 mins]

3. (U&F p 2.36) A typical vibration isolation mount consists of a rubber cylinder of diameter d compressed inside a steel cylinder by a force Q applied to a steel rod, as shown in figure P2.36. Find, in terms of d, Q, and modulus E and Poisson's ratio v for the rubber (as needed): (a) an expression for the lateral pressure p between the rubber and the steel cylinder and (b) the value (and units) of p for d = 50mm, E = 10 MPa, v = 0.3, and Q = 5 kN. *Assumptions:* 1. Friction is negligible; 2. The steel cylinder and rod are rigid (why is this a good assumption?).

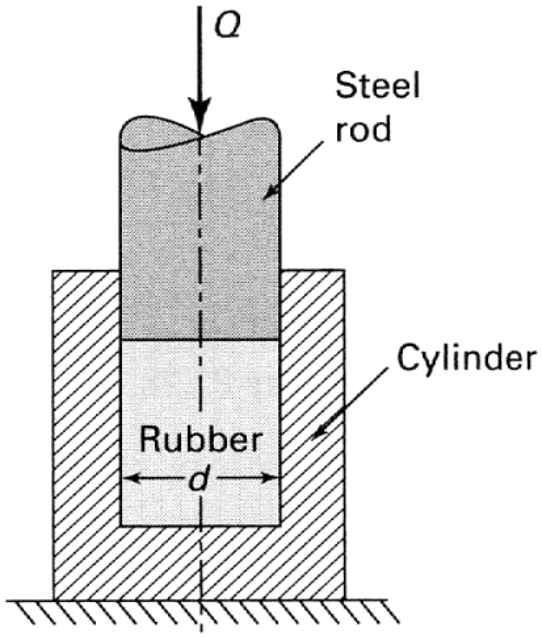


FIGURE P2.36.

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Approach to part (a):

- A stress element on the surface of the rubber cylinder would have the following state of stress:

$$\sigma_r, \sigma_\theta, \sigma_z$$

With zero shear because of the assumption that there is no friction

- The generalized Hooke's law for an isotropic homogeneous material and from the assumption that the steel cylinder is rigid and does not deform

$$\varepsilon_r = \frac{1}{E}[\sigma_r - \nu(\sigma_\theta + \sigma_z)] = 0, \qquad \varepsilon_\theta = \frac{1}{E}[\sigma_\theta - \nu(\sigma_r + \sigma_z)] = 0$$

Thus,

$$\sigma_r - \nu(\sigma_\theta + \sigma_z) = \sigma_\theta - \nu(\sigma_r + \sigma_z) \qquad (0 = 0)$$
$$\sigma_r - \nu\sigma_\theta = \sigma_\theta - \nu\sigma_r$$
$$(\sigma_r - \sigma_\theta) + \nu(\sigma_r - \sigma_\theta) = 0$$
$$(1 + \nu)(\sigma_r - \sigma_\theta) = 0$$

Then, we can say

$$\sigma_r = \sigma_\theta$$

Since there would be a compressive stress in the radial direction, let $\sigma_r = \sigma_\theta = -p$

- $\sigma_z = -\frac{Q}{A_{XC}}, \quad A_{XC} = \frac{\pi}{4}d^2$
- Now rewrite the generalized Hooke's law for p using ε_r or ε_θ since we know strain in those directions are zero

Approach to part (b)

- Substitute values into equation derived in part (a)

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Part A:

$$\tau_{r\theta} = \tau_{rz} = 0 \qquad \sigma_r = \sigma_\theta = -p \qquad \sigma_z = -\frac{4Q}{\pi d^2}$$

$$\varepsilon_r = \frac{1}{E}[\sigma_r - \nu(\sigma_\theta + \sigma_z)] = 0$$

$$\frac{1}{E}\left[-p - \nu\left(-p - \frac{4Q}{\pi d^2}\right)\right] = 0$$

$$\nu p + \frac{4\nu Q}{\pi d^2} = p$$

$$\frac{4\nu Q}{\pi d^2} = p(1-\nu) \qquad \boxed{p = \frac{4\nu Q}{(1-\nu)\pi d^2}}$$

Part B:

$$p = \frac{4\nu Q}{(1-\nu)\pi d^2} = \frac{4[0.3][5\times 10^3\ N]}{(1-[0.3])\pi[0.05\ m]^2}$$

$$\boxed{p = 1.091\times 10^6\ N/m^2}$$

The assumption that the steel cylinder and rod are rigid allows us to constrain the strain to be zero in this problem, allowing us to solve for the pressure exerted between the rubber and steel cylinder.
[Time spent: 35 mins]

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4. (Based on U&F p 2.42) The strain readings from a 60º strain gauge rosette mounted at point *A* on a loaded C-clamp (shown in Figure P2. 42) are $\varepsilon_a = 900\ \mu$, $\varepsilon_b = 340\ \mu$, and $\varepsilon_c = -80\ \mu$. (a) Find the magnitudes and directions of the principles strains. (b) Find the maximum shear strain and its direction.

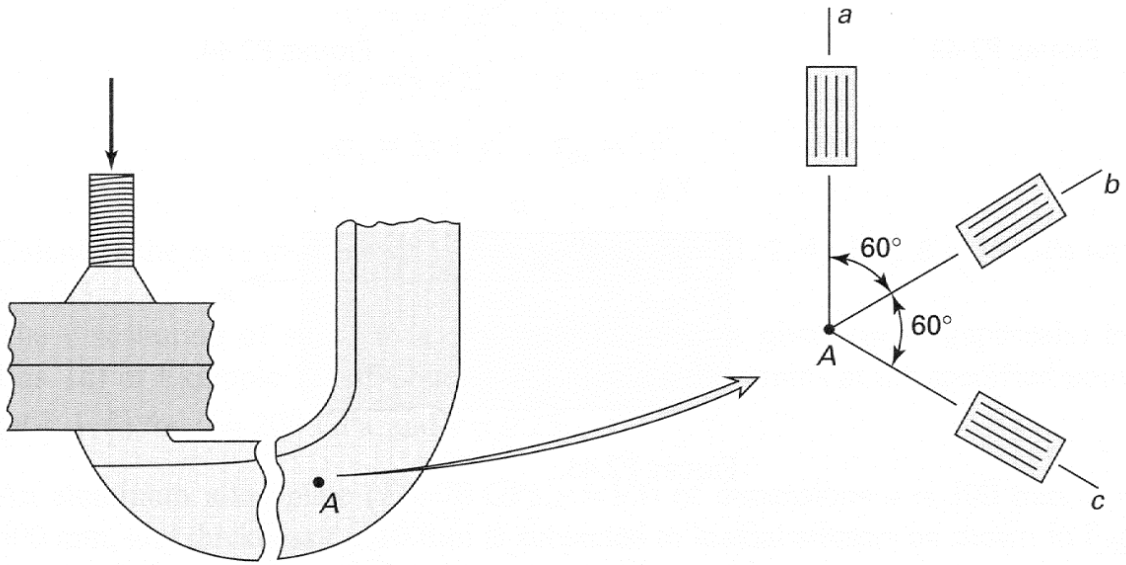


FIGURE P2.42.

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Approach to part (a):

- Setting the x-axis direction to be aligned with such that

$$\varepsilon_x = \varepsilon_a$$

- Using strain transformation formulas to solve for the state of strain (know ε_x, find ε_y and γ_{xy}). Letting

$$\varepsilon_{x'} = \varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta\,, \qquad \theta = -60^\circ$$

$$\varepsilon_{x''} = \varepsilon_c = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta\,, \qquad \theta = -120^\circ$$

- Knowing the state of strain, find the principal strains and directions using:

$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y}, \qquad \theta_{p1} + 90^\circ = \theta_{p2}$$

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

Approach to part (b):

- Similar to the Mohr's circle for stress, the max shear strain would be found 90 deg from the principal directions (corresponding to a 45 deg rotation physically). With its magnitude given by

$$\frac{\gamma_{max}}{2} = (radius\ of\ the\ Mohr's\ circle) = \frac{\varepsilon_1 - \varepsilon_2}{2}$$

$$\gamma_{max} = \varepsilon_1 - \varepsilon_2$$

$$\theta_s = \theta_{p1} \pm 45^\circ$$

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Part A:

$$\varepsilon_x = \varepsilon_a = 900 \times 10^{-6}$$

$$\varepsilon_{x'} = [340 \times 10^{-6}] = \frac{[900 \times 10^{-6}] + \varepsilon_y}{2} + \frac{[900 \times 10^{-6}] - \varepsilon_y}{2}\cos(2 \cdot -60°) + \frac{\gamma_{xy}}{2}\sin(2 \cdot -60°)$$

$$0.00045 + 0.5\varepsilon_y - 0.000225 + 0.25\varepsilon_y - 0.43301\gamma_{xy} = 0.00034$$

$$0.75\varepsilon_y - 0.43301\gamma_{xy} = 0.000115$$

$$\varepsilon_{x''} = [-80 \times 10^{-6}] = \frac{[900 \times 10^{-6}] + \varepsilon_y}{2} + \frac{[900 \times 10^{-6}] - \varepsilon_y}{2}\cos(2 \cdot -120°) + \frac{\gamma_{xy}}{2}\sin(2 \cdot -120°)$$

$$0.00045 + 0.5\varepsilon_y - 0.000225 + 0.25\varepsilon_y + 0.43301\gamma_{xy} = -0.00008$$

$$0.75\varepsilon_y + 0.43301\gamma_{xy} = -0.000305$$

$$\begin{aligned} & 0.75\varepsilon_y - 0.43301\gamma_{xy} = 0.000115 \\ + \quad & 0.75\varepsilon_y + 0.43301\gamma_{xy} = -0.000305 \\ = \quad & 1.5\varepsilon_y = -0.00019 \end{aligned}$$

$$\varepsilon_y = -0.000126$$

$$\begin{aligned} & 0.75\varepsilon_y - 0.43301\gamma_{xy} = 0.000115 \\ - \quad & 0.75\varepsilon_y + 0.43301\gamma_{xy} = -0.000305 \\ = \quad & -0.86602\gamma_{xy} = 0.00042 \end{aligned}$$

$$\gamma_{xy} = -0.000484$$

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$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} \longrightarrow 2\theta_p = \tan^{-1}\left(\frac{-0.000484}{0.0009 - -0.000126}\right) = -25.254°$$

Since the numerator is negative and denominator is positive, this corresponds to a point in the 4th quadrant, and thus

$\boxed{\theta_{p1} = -12.62°}$

$\boxed{\theta_{p2} = -12.62° + 90° = 77.37°}$

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} = \frac{[0.0009 + -0.000126]}{2} \pm \sqrt{\left(\frac{[0.0009 - -0.000126]}{2}\right)^2 + \left(\frac{[-0.000484]}{2}\right)^2}$$

$\varepsilon_{1,2} = 0.000387 \pm 0.000567$

$\boxed{\varepsilon_1 = 0.000954}$ $\boxed{\varepsilon_2 = -0.0001804}$

Part B:

$\gamma_{max} = \varepsilon_1 - \varepsilon_2$

$\boxed{\gamma_{max} = 0.00113}$

$\theta_s = \theta_{p1} \pm 45° = -12.62° \pm 45°$

$\boxed{\theta_{s1} = 32.38°}$ $\boxed{\theta_{s2} = -57.62°}$

The principal strains indicate that the material experiences greater tensile strain than compressive strain. The maximum engineering shear strain occurs at directions rotated 45° from the principal axes, consistent with the strain transformation equations. [Time spent: 30 mins]